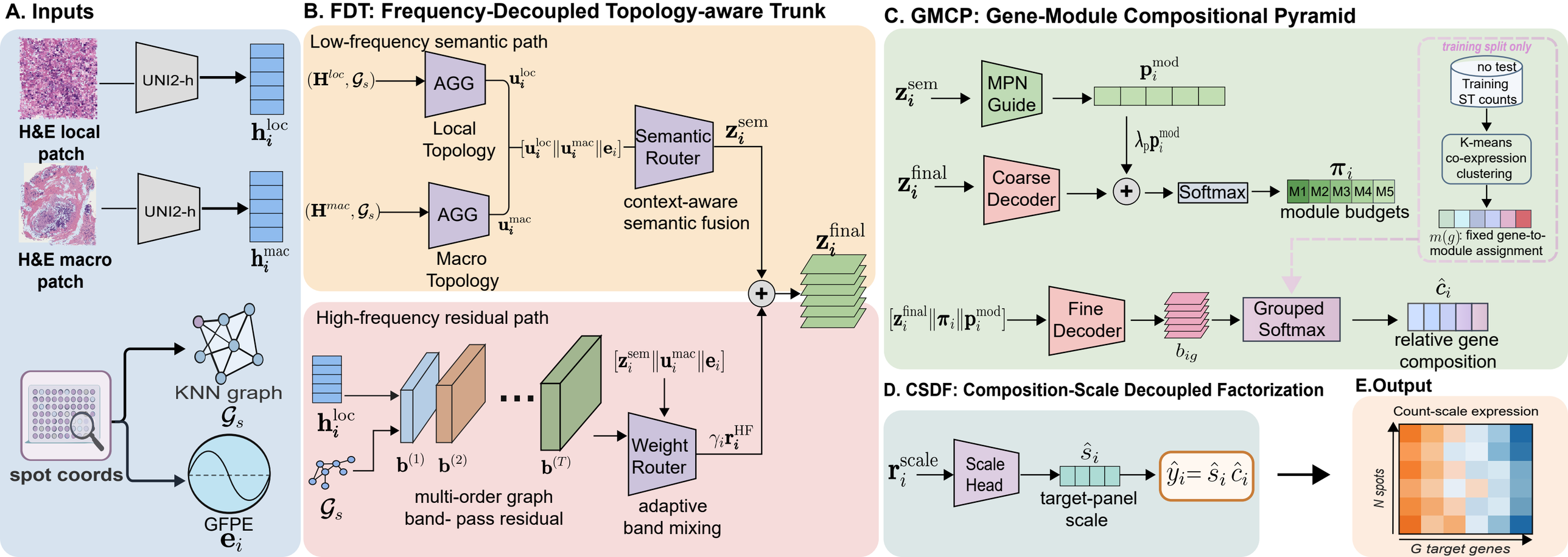

A. Inputs
H&E local patch
UNI2-h
$\mathbf{h}_i^{\text{loc}}$
H&E macro patch
UNI2-h
$\mathbf{h}_i^{\text{mac}}$
spot coords
KNN graph
$\mathcal{G}_s$
GFPE
$\mathbf{e}_i$
B. FDT: Frequency-Decoupled Topology-aware Trunk
Low-frequency semantic path
$(\mathbf{H}^{loc}, \mathcal{G}_s)$
AGG
Local Topology
$\mathbf{u}_i^{\text{loc}}$
$(\mathbf{H}^{mac}, \mathcal{G}_s)$
AGG
Macro Topology
$\mathbf{u}_i^{\text{mac}}$
$[\mathbf{u}_i^{\text{loc}} \| \mathbf{u}_i^{\text{mac}} \| \mathbf{e}_i]$
Semantic Router
context-aware semantic fusion
$\mathbf{z}_i^{\text{sem}}$
$\mathbf{z}_i^{\text{final}}$
High-frequency residual path
$\mathbf{h}_i^{\text{loc}}$
$\mathcal{G}_s$
$\mathbf{b}^{(1)}$
$\mathbf{b}^{(2)}$
$\mathbf{b}^{(T)}$
multi-order graph band- pass residual
$[\mathbf{z}_i^{\text{sem}} \| \mathbf{u}_i^{\text{mac}} \| \mathbf{e}_i]$
Weight Router
$\gamma_i \mathbf{r}_i^{\text{HF}}$
adaptive band mixing
C. GMCP: Gene-Module Compositional Pyramid
$\mathbf{z}_i^{\text{sem}}$
MPN Guide
$\mathbf{p}_i^{\text{mod}}$
$\lambda_p \mathbf{p}_i^{\text{mod}}$
$\mathbf{z}_i^{\text{final}}$
Coarse Decoder
Softmax
$\boldsymbol{\pi}_i$
M1 M2 M3 M4 M5
module budgets
training split only
no test
Training ST counts
K-means co-expression clustering
$m(g)$: fixed gene-to-module assignment
$[\mathbf{z}_i^{\text{final}} \| \boldsymbol{\pi}_i \| \mathbf{p}_i^{\text{mod}}]$
Fine Decoder
b_{ig}
Grouped Softmax
$\hat{c}_i$
relative gene composition
D. CSDF: Composition-Scale Decoupled Factorization
$\mathbf{r}_i^{\text{scale}}$
Scale Head
$\hat{s}_i$
target-panel scale
$\hat{y}_i = \hat{s}_i \hat{c}_i$
E.Output
Count-scale expression
N spots
G target genes